Zoning and the American Suburb*

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Abstract

We construct the first panel dataset of suburban zoning maps and bylaws for a major American metropolitan area using the setting of greater Chicago. Combining this dataset with detailed public records allows us to observe the regulatory environment under which land in the suburbs was subdivided and developed. Restrictive zoning emerged earlier than widely believed, and lot size regulations were binding as early as the 1920s. However, new residential neighborhoods were becoming less walkable, with fewer stores, in both zoned and unzoned municipalities. Using variation in the restrictiveness of zoning assigned to undeveloped land, we estimate suburban zoning's impact over the long run. Areas zoned more flexibly in the past have 14% more land for apartments and 16% more land for commercial uses today relative to areas zoned the most restrictively. Zoning substantially increased the prevalence of single-family residential neighborhoods relative to the suburban form that would have prevailed without such regulation, which would have featured more mixed uses and smaller lots.

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1 Introduction

American suburbs are notable for their residential uniformity and extreme separation of uses. Jane Jacobs wrote of suburban developments of the 1950s – composed almost exclusively of detached single-family homes – as exhibiting a remarkable degree of “monotony, sterility, and vulgarity” (Jacobs, 1961, p. 7). Why did compact neighborhoods with a mix of residential and commercial uses stop being built in American metropolitan areas? And why are so many American suburbs filled with single-family homes on large lots that are inaccessible except by car and unwalkable even to the nearest store?

While critics and fans alike have pointed to land use regulation as the explanation for the distinctive form of American suburbs, we know surprisingly little about how important these regulations have been over the long run. Upwardly mobile Americans increasingly sought homes in planned neighborhoods and viewed apartment buildings as “parasitic” nuisances to be avoided.[1] These preferences for neighborhoods complicate efforts to understand the role of zoning during the 1920–1980 period when a significant share of the U.S. housing stock was constructed.

The central challenge in making progress on the long-run impact of zoning on the suburbs is in observing the regulatory environment at the time that these places were originally developed. Historical regulations can be very difficult to locate, as they have nearly all been superseded and relegated to municipal archives and libraries. Accordingly, ambitious efforts to map the nation’s zoning laws such at the National Zoning Atlas (Xu et al., 2023) do not provide information on historical regulations, nor do widely used zoning surveys such as the Wharton Regulatory Land Use Regulation Index (Gyourko et al., 2008). The lack of information on historical zoning laws has limited efforts to understand the impact of zoning over the long run, particularly because current regulations have likely been shaped by many of the economic forces social scientists wish to study (Davidoff et al., 2016; Molloy, 2020; Trounstine, 2020).

In this paper we construct the first panel dataset of zoning regulations for a major U.S. metropolitan area using the original maps and bylaws from suburban municipalities around Chicago. The Cook County Longitudinal Database covers regulations from 1921, when suburban zoning first

[1]The original case legalizing comprehensive land use regulation Euclid v. Ambler (1926) contained the following description of apartments in suburban neighborhoods: “very often the apartment house is a mere parasite, constructed in order to take advantage of the open spaces and attractive surroundings created by the residential character of the district.”

appeared, to the 1970s. We combine these historical zoning regulations with unusually detailed spatial data that we obtained through a public records request to Cook County. This data includes a map with the date of subdivision for parcels and the date of construction of existing houses. This information is important because parcels were often subdivided decades before houses were built on them, limiting the usefulness of national datasets from providers such as CoreLogic that contain only structure built dates for studying lot sizes.[2] Cook county contains over 100 municipalities that developed at different times and under various regulatory environments, yielding a broad variety of contemporary suburban forms.

We use this dataset establish several new facts about early suburban zoning ordinances and residential development. Suburban ordinances set aside a striking amount of land for single-family residential uses. Among the 33 zoning ordinances adopted by inner-ring suburbs in the 1920s, 88% of land was reserved for single-family homes. The corresponding share was just 4% in the zoning ordinance adopted by Chicago in 1922. Meanwhile, required minimum lot sizes became substantially larger over time, from 6,200 square feet in the 1920s to 7,500 square feet in the 1970s. Because we can observe date of plat for every parcel in the county, we can observe whether development occurred under regulation or before it was adopted. We find that unzoned development dominated in the 1920s, trailed off after World War II, and disappeared almost completely by 1970. This timeline motivates the empirical work we undertake in the paper.

We begin our analysis by exploring how the introduction of land use regulation changed the character of residential neighborhoods. We find that, conditional on location within the county and year of development, new single-family residential neighborhoods were becoming less walkable and further from apartments and stores in both regulated and unregulated municipalities. Zoning explains little of the differences across residential neighborhoods in the prewar era. The one exception is minimum lot size (MLS) regulations. Although prevailing wisdom suggests that density restrictions were not important until the 1970s (for instance, see Fischel, 2005; Krimmel, 2021), we find that minimum lot sizes were binding as early as the 1920s. Zoning was associated with a 1,622 square foot increase in lot size (25% of the mean) in residential development from 1921 to 1945. We also compare development of different regulatory restrictiveness in the postwar era (1945–1980)

[2] For example, more than half of the homes built in suburban Cook county in the early 1950s were built on parcels that were subdivided in the 1920s or before.

and find an elasticity of lot sizes with respect to MLS of .47.

The robustness of these effects strongly suggests that zoning shifted the lot sizes of residential neighborhoods beyond what the private market would have delivered. But our descriptive work highlights that zoning may have shifted the overall distribution of land use, principally towards single-family homes and away from virtually everything else. To make progress on this question, we use the Cook County Longitudinal Database and the Public Land Use Survey System (PLSS) grid, which was created as part of the Land Ordinance of 1785, to measure the impact of zoning on land that was undeveloped at the time regulation was adopted. Using such "greenfield" development minimizes the chance that zoning followed existing land use. In addition, we can control for a rich set of attributes of the land in each grid square, including its estimated value and proximity to Chicago, the lake, and transportation networks.

We show areas that were developed before zoning was in place had on average 55% of their land set aside for single-family homes and 15% for each of apartments and commercial uses (with the remainder largely for industry). But under zoning this distribution shifted to 80% single-family homes and 5% for each of apartments and commercial uses. Using simple entropy indexes to measure the diversity of zoning and contemporary land use, we ask how zoning restrictiveness at the time of development affected the overall distribution of land use. Areas with more diverse zoning at the time that land was subdivided have 14% more land for apartments today and 16% more land for stores relative to areas that were zoned the most restrictively. We show that these estimates are essentially invariant to the the inclusion of controls for other factors that could drive these differences, including the location and value of land. These results together suggest that zoning regulations themselves shifted the allocation of land in the suburbs. We conclude that suburban development would have contained substantially more mixed-use development had comprehensive land use regulation not been permitted by *Euclid v. Ambler Realty Co.* in 1926.

The data work in this paper required a years-long archival effort and the existence of an unusually well-equipped county GIS office. It is extraordinarily difficult to replicate this approach for a broader sample of metro areas. We close by considering what we can learn from our setting, in which near everything is observed, to guide interpretation of more indirect approaches. A growing number of recent papers have developed creative methods to impute past minimum lot size requirements from contemporary real estate data (Cui, 2024; Macek, 2024; Song, 2025; Zabel & Dalton, 2011). These approaches are undoubtedly important, as zoning has become a crucial part of the national debate on housing. Yet our findings suggest the need for caution in interpreting today's lot size distribution as a measure of past regulation. We find that suburban developers chose lot sizes clustered around similar round numbers whether zoning was in place or not. Zoning ordinances also largely followed existing subdivision, so the distribution of lot sizes from unregulated development was crystallized into future minimum lot size regulations. The observed lot size distribution today is thus informative of past minimum lot size regulations both because these regulations shaped development, and because existing development shaped future regulations. The dataset we have assembled for this paper provides the ground truth for one metro area, which will provide a new means of validating approaches with broader geographic coverage.

This paper contributes to a literature in economics that seeks to understand the causal impacts of land use regulation. Economists have studied the short-run impact of zoning on undeveloped areas (Turner et al., 2014) and on existing urban development, both in recent years (Anagol et al., 2021) and in historical settings (McMillen & McDonald, 2002). This emphasis on identification complements the broader literature on land use regulation, which argues that zoning can be used to preserve local control and increase home values (Fischel, 2005) but at the cost of excluding less privileged households (Trounstine, 2018). The increase in home values is related to the strong preferences of U.S. households for low-density development (Gyourko & McCulloch, 2023). Historical zoning parameters beyond minimum lot sizes have received comparatively little attention in economics due to the difficulty of observing them. Two existing papers that study the long-run, causal impact of zoning focus on central cities only (Shertzer et al., 2018; Twinam, 2018).

The archival zoning ordinances digitized for this paper represent a new avenue of inquiry for the study of zoning. The current literature on zoning largely relies on surveys (Glickfeld & Levine, 1992; Gyourko et al., 2008), digitized maps of contemporary regulations (Xu et al., 2023), and promising new methods using natural language processing to analyze current zoning codes (Bartik et al., 2025). While all of these methods are useful for understanding the constraints on current development (for instance, Kulka et al., 2024), none of them allow researchers to observe the exact regulations in place when a particular parcel was developed. Our approach, which trades off geographic coverage for an exceptional level of detail, provides a new benchmark for understanding both the impact of zoning and the accuracy of our measures of regulation. However, imputation approaches are currently limited to lot size regulations, and developing methods to study allowable uses remains a key challenge for future research. Observing the date when lots were subdivided in broader settings is another challenge for future work, as the sizable gap between the date of plat (which is typically not observed) and the date of structure (which is easily observed from providers such as CoreLogic) we document in the Chicago suburbs is likely present in many other settings.

Zoning has been linked to persistent racial segregation, disparities in public goods provision, and accelerating regional inequality (Ganong & Shoag, 2017; Gyourko & Krimmel, 2021; Hsieh & Moretti, 2019; Sahn, 2021; Schuetz, 2022). Restrictive land use regulation is thus intertwined with some of the most critical socioeconomic challenges facing the United States. This paper contributes to this larger literature by clarifying the role of regulation in shaping the form of many suburban areas in the United States. Initial development is costly to change (Brooks & Lutz, 2019), so the decisions made in past decades will constrain development far into the future, complicating efforts to increase the density and affordability of housing in high-demand areas of the United States.

2 The Cook County Longitudinal Zoning Database

The data for this paper was constructed from archival zoning ordinances from the municipalities of suburban Cook County, the second most populous county in the country and home to the city of Chicago. Our panel of zoning regulations begins in 1921 with the first three suburbs to adopt zoning ordinances: Evanston, Glencoe, and Oak Park. Our panel ends in 1970. The zoning ordinances all have two components, the zoning district map and the associated bylaws. The maps and bylaws were typically archived separately, with maps in some libraries and the bylaws in others. We found the various ordinance components in academic libraries in Illinois and in public libraries, including the Library of Congress. In some cases we made public records requests to the municipal offices themselves. If these approaches did not work, we used Olcott's Blue Books of Chicago to fill in missing bylaw information.[3] All of the maps were digitized using the 2020 street from from the Cook County GIS office as a reference.[4]

Zoning maps and bylaws were updated infrequently, but bylaws changed more frequently than

[3]The Blue Books were occasionally missing the most up to date regulations, and for this reason we tried to go back to the primary sources if they could be found.

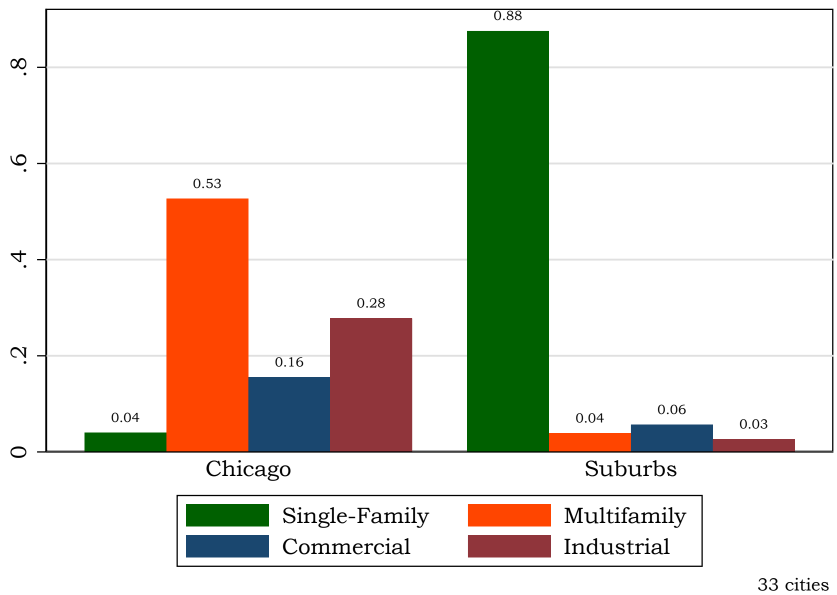
[4]https://hub-cookcountyil.opendata.arcgis.com/datasets/4569d77e6d004c0ea5fada54640189cf_5/explore.

the underlying map. We updated the bylaws each time they changed in a particular municipality for each year from 1921 to 1970. For the maps, we digitized the original zoning ordinance map adopted by a municipality in the 1920s if there was one. We then digitized updated or additional zoning ordinance maps at the start of each decade from 1940 to 1970. This approach allows us to assign the regulations in place when a parcel was divided with a high degree of accuracy. Because the maps and bylaws were changed infrequently, our panel describes the regulatory environmnent for most of the suburbs in the Cook county until well into the 1970s.

Cook County today contains 129 suburbs, 122 of which lie primarily within Cook County. Of these, 120 municipalities were incorporated and zoned by 1970. The coverage of the dataset expands each year as more suburbs incorporated and adopted zoning ordinances. In particular, the dataset covers 34 municipalities in the 1920s, 63 municipalities in 1940, 84 in 1950, 113 in 1960, and 111 in 1970. Cook County adopted a rural zoning ordinance in 1940 that applied to unincorporated land. Therefore, parcels developed on unincorporated land before 1940 were not subject to regulation. The rural ordinance had very large minimum lot sizes and was intended to deter development on unincorporated land (Erdmann, 1940). We focus on parcels developed in incorporated suburbs after zoning was adopted and parcels developed in unincorporated areas before 1940 or in incorporated suburbs before zoning was adopted.

To illustrate how we constructed the database for this paper, we reproduce a sample zoning ordinance map for Chicago Ridge in Appendix Figure A2a, along with its digitized counterpart in Appendix Figure A2b. This map is the zoning ordinance of 1945, which was still in place in 1950. The associated bylaws corresponding to each district can be seen in Appendix Figure A3. The various shading schemes show the allowable uses in each district. For instance, the double-hatched yellow area is reserved for "Heavy Industry," although the associated bylaws contain even more information, including the fact that residential uses were not explicitly banned. The two most restrictive districts allow only single-family homes but with different minimum lot sizes, 7,500 and 5,000 square feet. Appendix Figure A2c zooms in on the Chicago Ridge map to highlight the variation in zoning districts present, while Appendix Figure A2d shows that if we remove the zoning overlay, we can see from a contemporary aerial survey photo (circa 1938–1939) that the underlying area was largely vacant and undeveloped. The abundance of zoned but undeveloped land in our sample is an important part of our identification strategy discussed below.

Figure 1: 1924 Chicago zoning vs. early suburban zoning (zoned in 1920s)

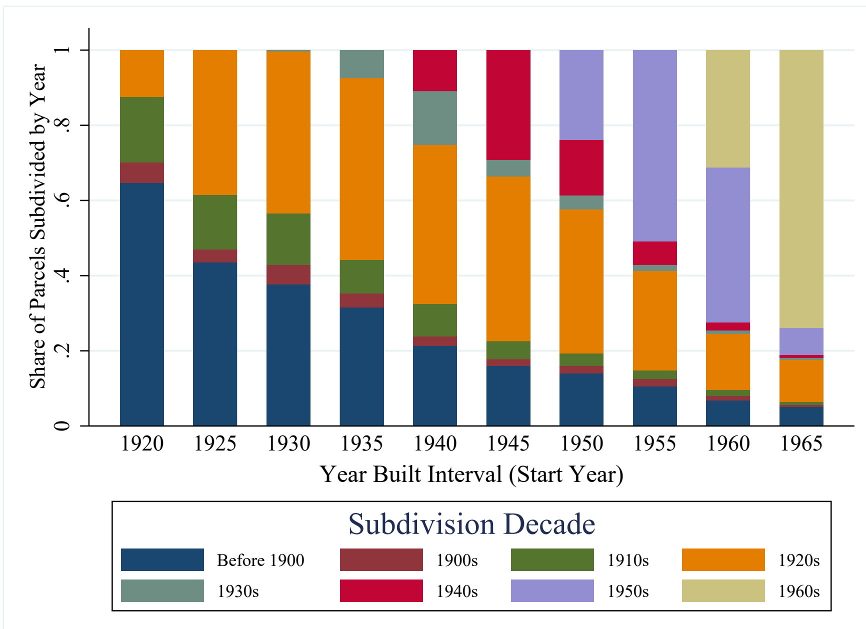


Note: The sample of suburbs includes all 33 Cook County municipalities with use zoning restrictions adopted in the 1920s.

Combining the various ordinances gives us an unprecedented view into the regulatory environment that shaped the development of the suburbs. Figure 1 shows the median use shares for the 33 suburbs in our sample that adopted zoning ordinances in the 1920s, compared with the initial zoning ordinance adopted by the city of Chicago in 1924. The suburban municipalities set aside a striking 88% of their land for single-family homes. Just 4% of land was zoned for apartments and 6% for commercial uses. The development allowed under these ordinances represents a dramatic break from existing urban form. Meanwhile, residential districts permitting apartments covered almost half of Chicago's land, with commercial areas allowing for mixed uses permitted on another 16%. We would expect some difference in use shares because the suburbs in this sample are further from the central business district of the city. We turn to the question of whether such proximity can explain the observed differences in Section 4.

We rely on several other data sources to construct our parcel-level dataset. National datasets do not typically contain information on when parcels were platted. However, we obtained subdivision plat maps for the suburbs, along with detailed subdivision information, through a public records request to Cook County. This data is crucial for linking each parcel to the regulatory environment under which it was developed, because the date on which any structure was built on the parcel can be much later than the date on which the underlying land was subdivided. Figure 2 shows the gap between the year a parcel was subdivided and the year a structure was built on it in five-year intervals over our sample period. The average gap between year built and year subdivided across our whole sample is 21 years. For instance more than half of the homes built in suburban Cook county in the early 1950s were built on parcels that were subdivided in the 1920s or before, underscoring the importance of using year of plat rather than the year built for of current structure on the parcel.

Figure 2: Gap between Parcel Plat and Structure Built Year



Note: This figures shows the parcel plat distribution for houses built over five-year intervals. For example, the 1920 bar contains single-family houses built from 1920 to 1924, the 1925 bar from 1925 to 1929, and so on.

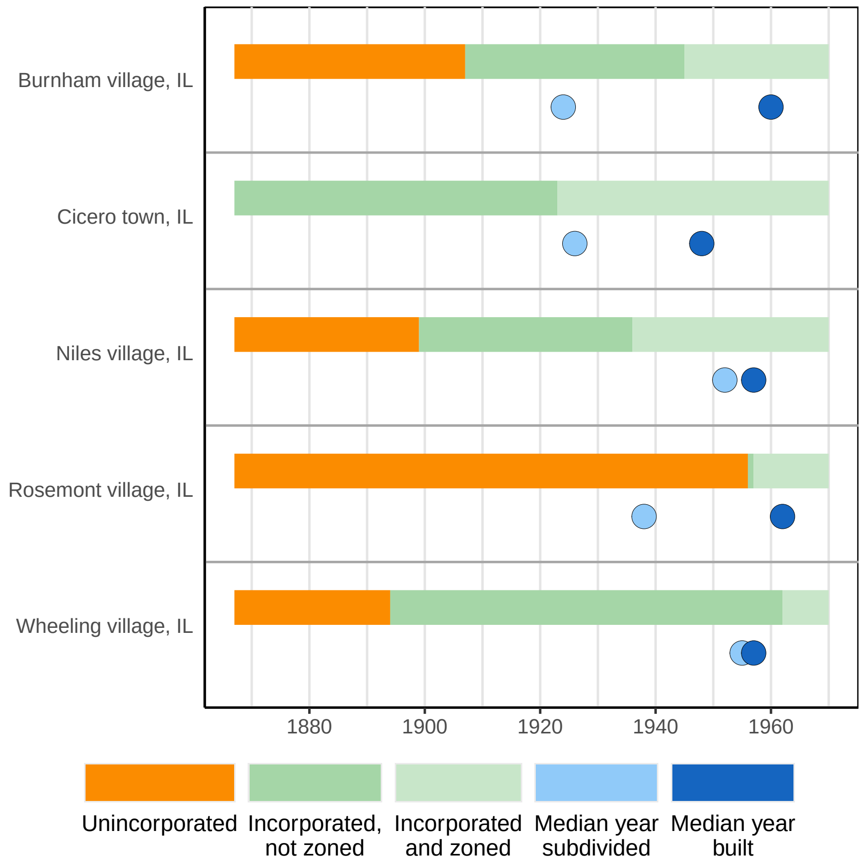
We use several other data sources in our analysis. We obtained information on when each parcel became part of an incorporated municipality from the Cook County GIS Office's Incorporation Inventory.[5] This allows us to determine if parcels were subject to the regulations adopted in that suburb at time of subdivision or development. The parcel map for Cook County was also obtained through the Cook County GIS Office.[6] We obtained information on the location of present-day

[5]Cook County Municipal Incorporation Inventory, accessed 2023. See https://maps.cookcountyil.gov/mii/.

[6]Cook County GIS. Historical Lots, 2017 (updated August 2022). See https://hub-cookcountyil.opendata.arcgis.com/datasets/2d2d5e9fb3934612809109b6833e5897_17/explore.

uses, including apartments and stores, from the Chicago Metropolitan Agency for Planning.[7]

Figure 3: Development Timeline



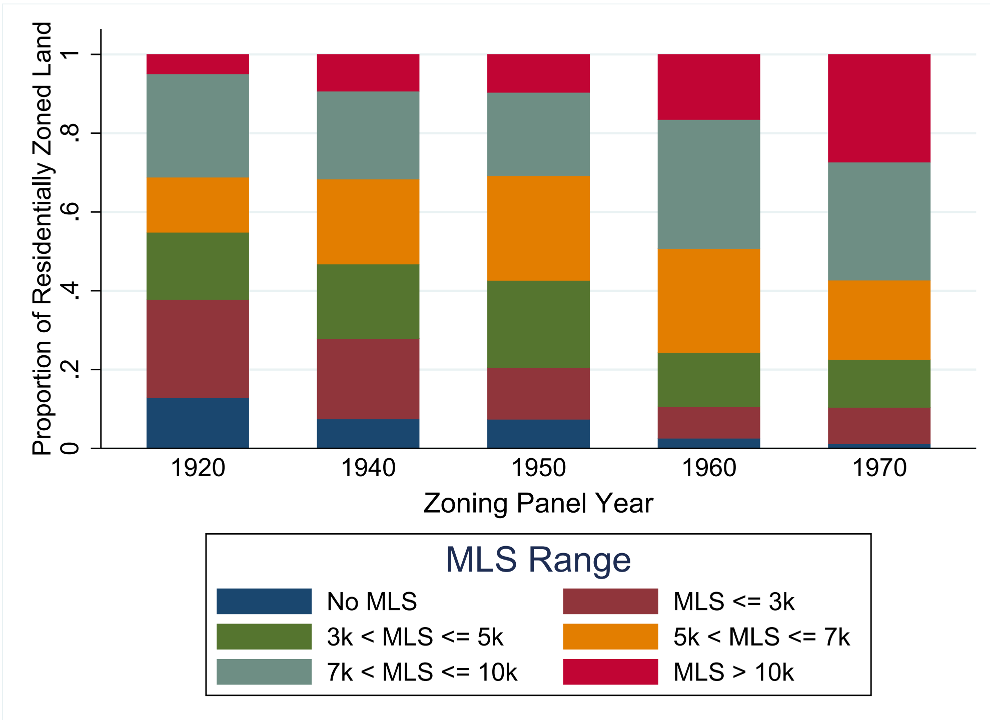
Note: This figure depicts the timeline of incorporation and zoning for five cities in our sample over the period 1867-1970. Cicero was the first of these cities to incorporate in 1867. Also depicted are the median year subdivided and median year built for residential parcels in each city.

This combined dataset contains rich variation in the regulatory environment under which suburbs were developed. Figure 3 illustrates the timing around development and regulation in Cook County for five representative municipalities, along with the median year that parcels were subdivided and houses were built. The inner ring suburb of Cicero was substantially built up by the time that zoning was adopted in December 1923, with about half of parcels subdivided after Cicero was incorporated but before it was zoned. On the other hand, much of Rosemont was subdivided before it was incorporated in the mid-1950s. Burnham Village and Niles Village adopted zoning around 1940, but Burnham Village had substantially more development before it was zoned relative

[7]The 2018 Land Use Inventory for Northeastern Illinois can be found at https://datahub.cmap.illinois.gov/datasets/ef5f2092556c409d94d119bd821ec6da_2/explore?location=41.841836%2C-88.109641%2C9.13.

to Niles Village. The dataset we have assembled allows us to ask how, conditional on the year and location within the county, regulation shaped suburban development.

Figure 4: Distribution of Minimum Lot Size Restrictions by Year



Note: This figure summarizes the distribution of minimum lot size (MLS) required per household by year. Zoning district observations are weighted by land area. N = 2,353.

We also visualize the increasing restrictiveness of zoning ordinances over time using the easily comparable parameter of minimum lot sizes in Figure 4. This is the first visualization of the actual minimum lot sizes that were in place by *zoning district* over this period. It is worth noting that survey-based approaches to collect zoning information typically ask about the most restrictive lot size restriction in the municipality. With our data we can see the whole range of density restrictions that applied in different zoning districts (we weight zoning districts by land area for representativeness). The median lot size requirement was 6,200 square feet in the 1920s and grew to 7,500 in the 1970s. We discuss below how lot sizes were often much larger than 7,500 square feet by the end of our sample period, underscoring the importance of having independent measures of regulation.

In our empirical work below, we also include covariates that measure the distance between our unit of observation and Lake Michigan, the nearest major river, the central business district of

Chicago, the city boundary of Chicago, and the nearest 1942 zoning use district in Chicago (single-family, townhome, apartment, commercial, and industrial) as well as the value of land in 1943. These controls have various sources.[8] For our analysis of zoning restrictiveness across ordinances, we rely on the division of Cook County into one-mile sections established by the Land Ordinance of 1785, which laid the ground for the Public Land Survey System (PLSS). We obtained the map of PLSS sections from the Cook County GIS office.[9] Finally, Walkscores were from accessed from Walkscore.com (as of 2023).

3 Early Suburban Development and Trends

Zoning represents a formalization of demand for controlled development of low-density, purely residential neighborhoods, which became feasible on a larger scale with the arrival of the private automobile. However, this same technology allowed motor trucks to become widespread over the 1920s, heightening concerns about the arrival of commercial and industrial enterprises in suburban neighborhoods (Fischel, 2004). The concern about mixed uses was not misplaced, as in addition to the quality of life considerations, purely residential zoning was associated with higher home values after World War I (McMillen & McDonald, 2002). Homeowners therefore placed a premium on being able to control development near their their homes, both for reasons of property values and aesthetics. The decision in *Euclid v. Ambler*, which allowed comprehensive zoning to stand, provided a new avenue for control of land use at the municipal level.

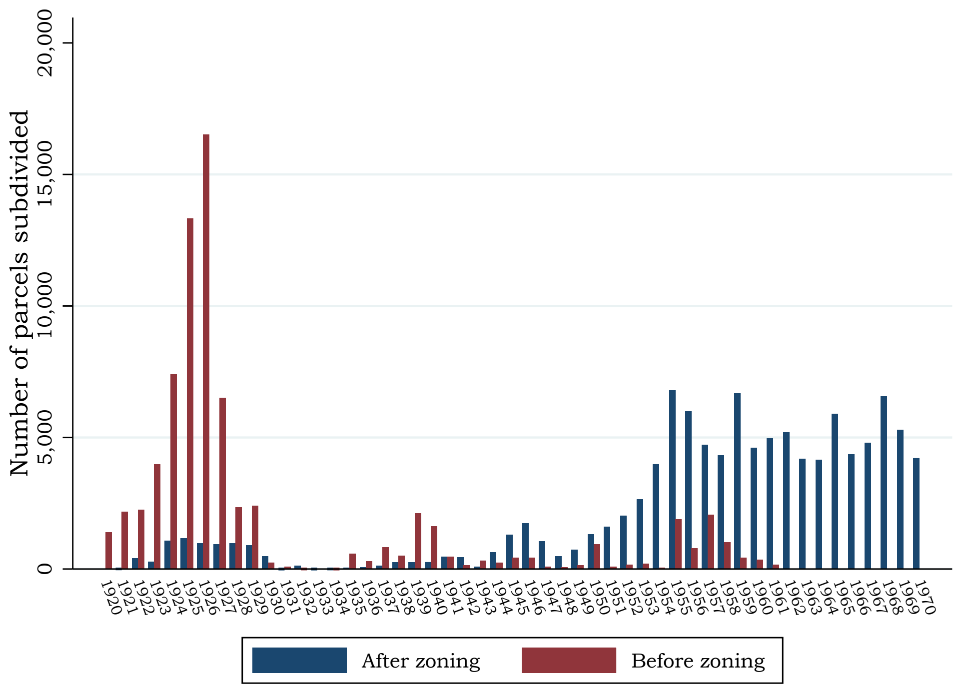
But not all residential uses were created equal. The text from the *Euclid v. Ambler* case demonstrates that apartment buildings were viewed as social ills as early as the 1920s. Although the infamous language from the *Euclid* decision describing apartments as "parasitic" focuses on aesthetic concerns, other critics have argued that apartment buildings free ride in terms of property taxes. The notion that occupants of apartments pay less than owners of detached houses to access

[8] Cook County and the city of Chicago keep a wealth of geographic information online. We constructed the distance to the lake or river measure from https://hub-cookcountyil.opendata.arcgis.com/datasets/078a8acf872548eb9c91d68658f2895a_8/explore, the distance to city measure from https://data.cityofchicago.org/Facilities-Geographic-Boundaries/Boundaries-City/ewy2-6yfk, the distance to the CBD from https://data.cityofchicago.org/Facilities-Geographic-Boundaries/Boundaries-Central-Business-District/tksj-nvsw, and the distance to the nearest use district from our own digitization of Chicago's 1942 zoning ordinance. The source for this ordinance is the Chicago Planning Commission, 1942 Map of Zoning Districts of Chicago. The source of land values is Olcott's.

[9] Cook County GIS (2022), Public Land Survey Section, see https://hub-cookcountyil.opendata.arcgis.com/datasets/217a635972fb4dfa95411e57a57d1250_3/explore.

local public services appears frequently in the real estate literature over the past century, although some of this concern may have been misplaced.[10] The hostility to apartment buildings evident by the 1920s is part of a longer history of distaste for dense housing development, which was associated with overcrowding and disease (Hirt, 2015, p. 117). Indeed, developers used restrictive covenants to block apartment buildings in the urban periphery even in the 19th century, before comprehensive zoning regulations were introduced (Wolf, 2008).

Figure 5: Parcel development by year and regulation



Note: This figure depicts the timeline of parcel subdivision for suburban Cook County by year and regulatory status.

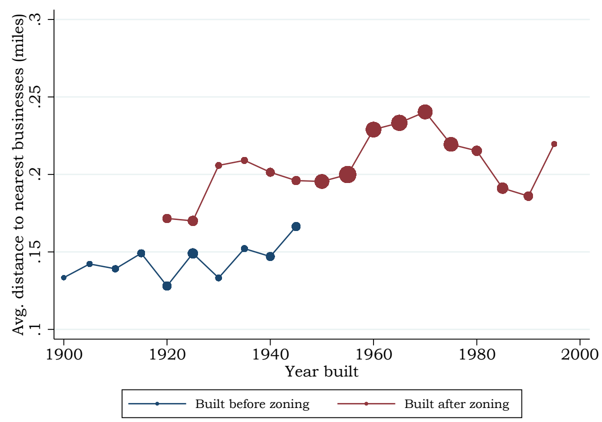
Given these strong preferences for "orderly" residential development composed entirely of single-family homes, one may expect the market to have delivered these kinds of neighborhoods even without zoning. It has previously been difficult to disentangle the role of regulation from the market because of the impossibility of observing historical regulations. Using our new dataset, we can visualize development by whether it was zoned or not over time. Figure 5 shows that unzoned development dominated in the 1920s, trailed off after World War II, and disappeared completely by

[10]For instance Gallagher (2016) finds that apartment buildings reduce property tax rates to achieve particular per-capita public spending levels in more recent decades.

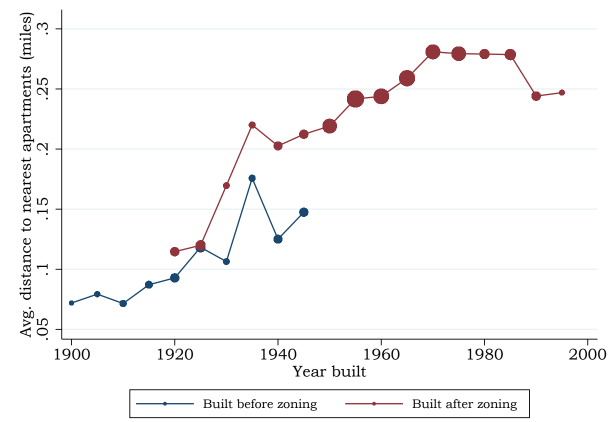
1970.[11] This timeline also motivates the empirical exercises we undertake in the rest of the paper.

4 The Impact of Zoning on Residential Neighborhoods

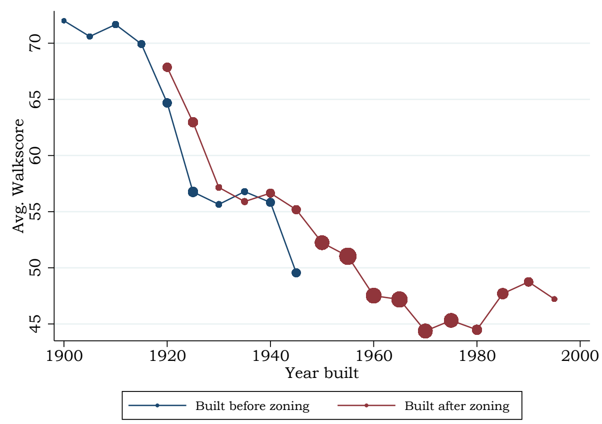
Figure 6: Trends in Suburban Single-Family Home Characteristics by Year Built



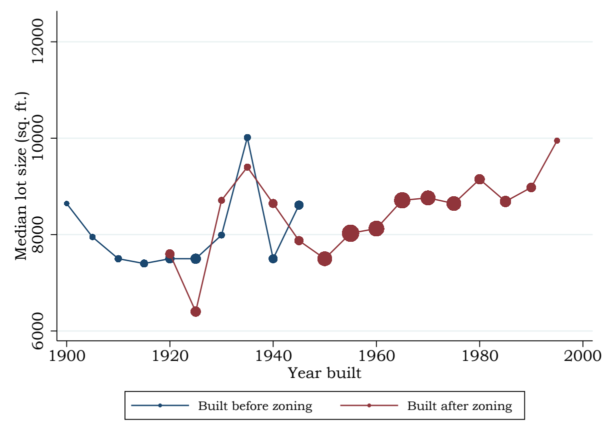
(a) Proximity to commercial



(b) Proximity to nearest apartment building



(c) Walkability



(d) Lot size

Note: Panel (a) and (b) depict the evolution (by year built) of the median distance to nearest commercial and apartment use, respectively, for single-family homes in suburban Cook County municipalities by whether they were built before or after zoning was adopted. Commercial and apartment uses are taken from a 2018 survey by the Chicago Metropolitan Agency for Planning. Panel (c) depicts the median Walkscore for parcels by year built; this metric was taken from Walkscore.com. Panel (d) depicts the evolution of mean lot sizes for single family homes by year built (observed in 2017). Dot sizes are scaled by the number of parcels represented.

To understand the impact of zoning on residential neighborhoods, we need a strategy that

[11]This figure also confirms the 1920s subdivision boom described by Homer Hoyt in his study of Chicago's development (Hoyt, 1933).

allows us to disentangle the role of regulation from broader changes in new neighborhoods being built over time, particularly as the development frontier moved further from Chicago. To visualize these trends, in Figure 6 we plot the characteristics of single-family homes in suburban Cook County today by the year in which they were built, separately for homes build under zoning or before it was adopted. The trends in commercial proximity in Figure 6a suggest that zoned residential development was generally further from the nearest store or retail establishment relative to single-family homes built in unzoned municipalities. However, the difference in average distance is about .06 miles, or 300 feet, which is not substantial, and both zoned and unzoned single-family home developments were gradually being built further from commercial uses.

We can also see that single-family homes were being built further from apartment buildings in both zoned and unzoned municipalities by the mid-1920s in Figure 6b. We similarly see declining walkability in the residential development in both zoned and unzoned municipalities in Figure 6c. The trends are very similar for this outcome, underscoring the need to also consider market explanations for changing development patterns when assessing the role of zoning. Figure 6d shows the trends in lot size across both regulated and unregulated development, which appear follow each other relatively closely. Lot sizes continue to expand after World War II as unzoned development trailed off (see Figure 5).

These figures suggest a surprisingly limited role for the impact of regulation on residential neighborhoods, with broader market forces shaping the character of new neighborhoods whether regulations were in place or not. To explore this idea further, we consider the universe of residential subdivisions built in Cook County from 1921 (the first year of any suburban zoning ordinance) to 1980 (the last year we treat as covered by the 1970 zoning ordinance) by their regulatory status at the time of development, which we continue to define as the date that the land was subdivided. Table 1 contains summary statistics for these subdivisions. Starting at the top of the table, it is clear that zoning increased the average lot size and reduced the coefficient of variation of lot sizes within subdivisions. Lots subdivided under zoning had a median size of 10,861 square feet, while those subdivided before zoning were just 8,267 square feet. Lots subdivided *and* built up before zoning had a median size of 9,441 square feet, suggesting the potential for compositional effects. Table 1 also shows that subdivisions developed under zoning were further from the nearest commercial use (.23 versus .16 miles) and apartment building (.27 versus versus .17 miles) and less walkable.

Table 1: Summary Statistics for Subdivisions, 1921-1980

	Subdivided after zoning	Subdivided before zoning	Subdivided & built before zoning
Actual lot size:			
▷ Average	10,861	8,267	9,441
▷ Median	8,938	6,704	7,669
CV of lot size:			
▷ Average	0.13	0.21	0.20
▷ Median	0.10	0.20	0.18
Dist. to commercial use in 2018:			
▷ Average	0.23	0.16	0.14
▷ Median	0.20	0.13	0.11
Dist. to apartments in 2018:			
▷ Average	0.27	0.17	0.16
▷ Median	0.21	0.13	0.10
Walkscore in 2023:			
▷ Average	47	55	57
▷ Median	48	56	58
Total number	6,314	1,181	311

Note: Summary statistics for residential subdivisions in Cook County; subdivisions contain an average of 27 individual parcels. Includes only subdivisions that are eventually incorporated or annexed into a suburban municipality, and excludes any subdivisions that were created on unincorporated land after the Cook County zoning ordinance of 1940 took effect. The "subdivided after zoning" column includes only subdivisions for which every parcel was subdivided after all parcels were subject to zoning. The "subdivided before zoning" column includes only subdivisions for which every parcel was subdivided before any were subject to zoning. The "subdivided & built before zoning" column adds the additional restriction to column 2 that 75% of parcels must have had homes constructed on them before being subject to zoning.

These summary measures, while suggestive, conflate zoning with other factors that were changing residential development patterns over time. Our empirical approach aims to identify the impact of zoning separately other attributes of new residential subdivisions such as the year of development and its location within Cook county. We first estimate the impact of zoning without controls using simple regressions relating subdivision characteristics to regulatory status at the time of development. The top panel of Table 2 shows the results of this regression with the following outcomes: subdivision average lot size, the coefficient of variation of lot size within the subdivision, average distance to the nearest commercial use, average distance to the nearest apartment building, and average Walkscore. Most of the effects are sizeable and statistically significant (with the exception of distance to nearest apartments and Walkscore).

Table 2: Zoning and Prewar (1921-1945) Subdivision Characteristics

	Avg. lot size (1)	CV lot size (2)	Dist. to com (3)	Dist. to apt (4)	Dist. to ind (5)	Avg. Walkscore (6)
Subdivided after zoning	1,698.60**	-0.02*	0.04**	0.02	0.23***	-0.47
	(633.80)	(0.01)	(0.02)	(0.02)	(0.06)	(1.79)
Observations	595	580	595	595	595	595
R^2	0.03	0.01	0.03	0.00	0.06	0.00
Controls	No	No	No	No	No	No
Subdivided after zoning	1,636.70***	-0.04***	0.04*	0.02	0.11**	-0.71
	(534.94)	(0.01)	(0.03)	(0.02)	(0.04)	(1.47)
Observations	588	574	588	588	588	588
R^2	0.35	0.18	0.22	0.23	0.52	0.32
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Note: Unit of observation is a subdivision, which may contain multiple parcels. In both panels, we restrict to subdivisions platted between 1921 and 1945. In panel the top panel, there are no additional covariates besides an indicator for whether or not the subdivision occurred under a zoning law. In the lower panel, we additionally include covariates for distances to the CBD of Chicago, the boundary of Chicago, the nearest of each zoning use type in Chicago in 1942, Lake Michigan, and the nearest river, as well as indicators for the year that subdivision began (which is typically the year most parcels were subdivided within each subdivision). Standard errors are clustered at year of subdivision.

In the lower panel of Table 2 we repeat this exercise controlling for other observable characteristics of parcels, namely the year subdivision occurred as well as a vector of relevant distance variables. These include distance to the central business district of Chicago, distance to the boundary of Chicago, distance to the nearest of each zoning use type in Chicago in 1942 (industrial, commercial, apartment, townhome, and single family residential), distance to Lake Michigan, distance to the nearest major river, and the value of land as measured in 1943. These controls are the most important determinants of the kind of residential development that would be delivered by market forces. The estimated effect of zoning is remarkably robust to the inclusion of controls in the regression, particularly for average lot size in the subdivision. Having a zoning ordinance in place was associated with average lot sizes that were 1,636 square feet larger relative to subdivisions in unzoned municipalities, an increase of 25% relative to the mean. The effects of zoning on other aspects of prewar subdivisions were small however, with a zoning ordinance associated with just a .04 mile (200 feet) further distance to the nearest store. These results suggest that zoning explains little of the changes in the built environment of prewar neighborhoods beyond lot size.

Table 3: Zoning and Postwar (1945–1980) Subdivision Characteristics

	Log avg. lot size (1)	Log dist. to com (2)	Log dist. to apt (3)	Log dist. to ind (4)	Avg. Walkscore (5)
Log avg. MLS	0.59***	0.61***	0.78***	0.68***	-16.97***
	(0.03)	(0.05)	(0.06)	(0.06)	(0.97)
Observations	5,500	5,493	5,420	5,500	5,500
R^2	0.19	0.05	0.06	0.06	0.10
Controls	No	No	No	No	No
Log avg. MLS	0.47***	0.31***	0.39***	0.34***	-7.84***
	(0.04)	(0.05)	(0.05)	(0.07)	(0.94)
Observations	5,465	5,458	5,385	5,465	5,465
R^2	0.29	0.12	0.17	0.28	0.26
Controls	Yes	Yes	Yes	Yes	Yes

Note: Unit of observation is a subdivision, which may contain multiple parcels. In both panels, we restrict to subdivisions platted after 1945 (inclusive) and subject to a minimum lot size. We exclude subdivisions with a greater than 15,000 square foot MLS (top 5% of the sample), as these areas were significantly more rural in character. In the top panel, there are no additional covariates besides the natural log of average MLS. In the lower panel, we additionally include covariates for distances to the CBD of Chicago, the boundary of Chicago, the nearest of each zoning use type in Chicago in 1942, Lake Michigan, and the nearest river, as well as indicators for the year that subdivision began (which is typically the year most/all parcels were subdivided within each subdivision). Standard errors are clustered at year of subdivision.

We also explore the intensive margin impact of zoning in the postwar era, when unzoned development was gradually disappearing. Specifically, we focus on the relationship between minimum lot size regulations and the built environment from 1945 to 1980 in municipalities that adopted comprehensive zoning ordinances. In Table 3 we report the relationship between the log of average MLS with respect to the same set of characteristics of the built environment, again without (top) and with (lower) controls for year of development and location within Cook county. The inclusion of controls reduces the estimated elasticity of average MLS with with respect to average lot size from .59 to .47. The relationship between log average MLS and the other characteristics of the built environment is reduced by about half with controls. As we may expect, larger MLS requirements affected other aspects of the built environment. Doubling the MLS reduces the Walkscore by 8 points (the average Walkscore for this sample is 48). Stores and apartments were also on average further away from single-family homes with larger MLS requirements.

Taken together, the robustness and magnitude of the effect on lot size strongly suggests that zoning impacted the built environment of suburbs beyond what we would have expected from the expanding urban frontier and forces in the private market. However, both zoned and unzoned developing were become less diverse and walkable as early as the 1920s. These results underscore the powerful incentives faced by developers to build detached single-family homes in exclusively residential neighborhoods.

5 The Impact of Zoning on the Distribution of Suburban Land Use

Our empirical work has thus far followed the existing empirical literature and focused on minimum lot size regulations and residential neighborhoods. But the descriptive analysis of the historical zoning ordinances suggests that these regulations may have been even more important for changing overall the distribution of land use in the suburbs, specifically towards single-family homes and away from everything else (see Figure 1). To answer this question, we need to move beyond residential subdivisions. We proceed using the grid produced by the Public Land Survey System (PLSS).

Championed by Thomas Jefferson, the Land Ordinance of 1785 that created the PLSS was intended to manage the subdivison and sale of government lands west of the Appalachian Mountains (Gates, 1996). Importantly for our application, the initial subdivision of land into six-mile-wide squares (townships) was done by surveyors before the earliest known non-Indigenous settlement in the area that would become Chicago. These townships were subsequently divided into one-mile-wide squares and then made available to potential settlers (Hirt, 2015); these were further divided into quarter-mile sections (the second division); this is the grid we use as our unit of observation.

We further focus on greenfield development to minimize the chance that the initial zoning classification of land followed existing uses (Shertzer et al., 2018). To measure the diversity of subsequent land use and zoning, we use simple entropy indexes. For each quarter-section square, we calculate shares zoned for single-family residential, apartment, commercial, and industrial use

Figure 7: PLSS Illustration

(a) Entropy of post-zoning land use (low diversity)

(b) Entropy of post-zoning land use (high diversity)

This figure provides a aerial view of the land use patterns of a quarter-section developed under zoning with low land use diversity in (a) and high land use diversity in (b).

in the earliest decade for which we observe zoning. Diversity is given by:

$$\sum_{i=1}^{N} s_i \ln(\frac{1}{s_i})$$

where N is the number of categories and s_i is the share of land devoted to zoning type i in the given decade. We also calculated shares devoted to actual land uses today using the same categories, plus a fifth category for townhomes, using the 2018 land use inventory from the Chicago Metropolitan Agency for Planning. The entropy index for zoning thus ranges from 0 to 1.35 and the entropy of land use today from 0 to 1.5.

To visualize the kind of variation in the data, Figure 7 provides an aerial view of two different quarter-mile sections. Figure 7a shows a square of land contained only single-family homes and a small industrial area. This is low diversity of land use in our data. In contrast, Figure 7b contains a broad mix of single-family homes, apartments, commercial uses, and industry. This is a high diversity of land use.

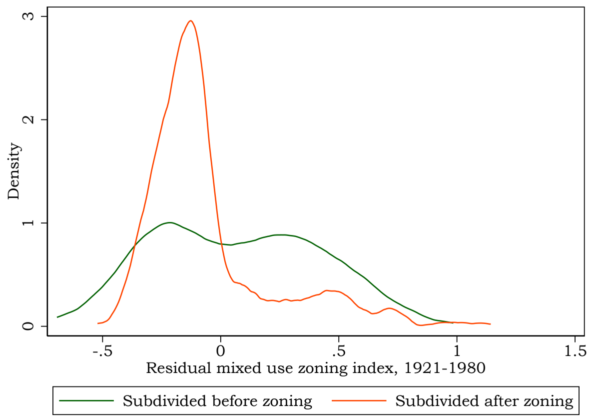
Broadly speaking, land developed under zoning has a very different distribution from land that was developed prior to the adoption of such regulations. Across Cook county, the average single-family/apartment/commercial/industrial mix if zoning came before development was 80%/5%/5%/10%. But the respective land use distribution that emerged from unregulated development was 55%/15%/15%/15%. This is a strikingly different distribution of land use, particularly away from apartments and stores and towards single-family homes. The question we address in the remainder of this section is whether this difference arose because of zoning or because of other factors specific to the timing and location of land that was developed after regulations were in place.

To provide the first piece of evidence on this point, we compare development patterns of quarter-sections subdivided under zoning compared with those subdivided after zoning, residualizing for year of development, 1943 land value, and controls for the quarter-section's location (the square's distance to the central business district of Chicago, distance to the boundary of Chicago, distance to the nearest of each zoning use type in Chicago in 1942 (industrial, commercial, apartment, townhome, and single family residential), distance to Lake Michigan, distance to the nearest major river). The residualized density plots are visualized in Figure 8.

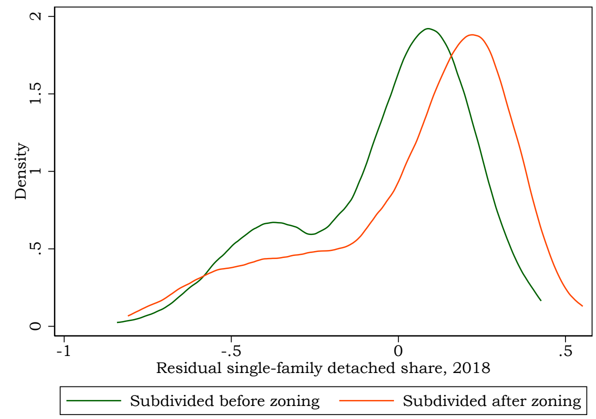
These density plots broadly suggest that that zoning had substantial impacts beyond what would have been predicted by market forces. In particular, Figure 8a shows a remarkable concentration of low-diversity zoning for mixed uses instead of the flatter distribution that prevailed before zoning. The decline in the diversity of zoning translated directly into a higher share of land reserved for single-family homes (Figure 8b) and a lower share of land reserved for commercial uses (Figure 8c). Collectively, we can see that the overall land use entropy distribution shifted to the left, representing a reduction in the diversity of land use today, relative to the distribution of land use entropy for quarter-sections subdivided before zoning (Figure 8d).

We again consider both the extensive and intensive margin impacts of zoning to reflect the fact that unregulated development became less common after World War II. Specifically, we restrict

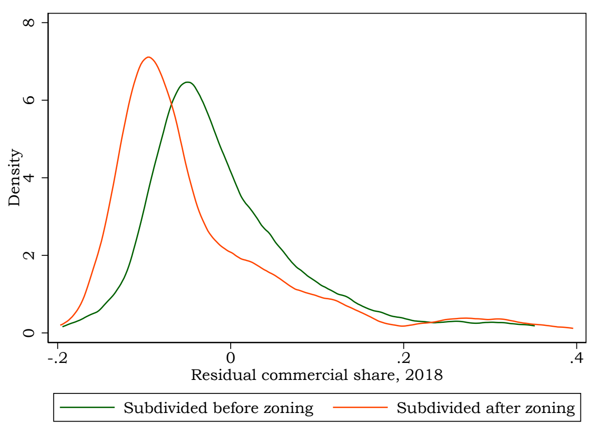
Figure 8: Impact of Zoning on Future Land Use Distribution



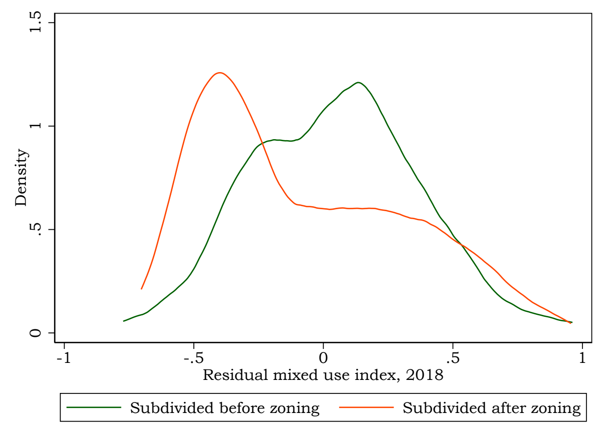
(a) Mixed use zoning index



(b) Share of land zoned for single-family homes



(c) Share of land zoned commercial



(d) Mixed land use index

Note: The outcome variable in each figure is residualized by the square's distance to the central business district of Chicago, distance to the boundary of Chicago, distance to the nearest of each zoning use type in Chicago in 1942 (industrial, commercial, apartment, townhome, and single family residential), distance to Lake Michigan, distance to the nearest major river, 1943 land value, and indicators for the median decade of development for parcels inside the square.

our attention to quarter-sections that were only developed – that is, subdivided into parcels – after zoning regulations were in place. We then run the following regression:

$$y_i = \alpha + \beta z_i + \gamma' X + \epsilon,$$

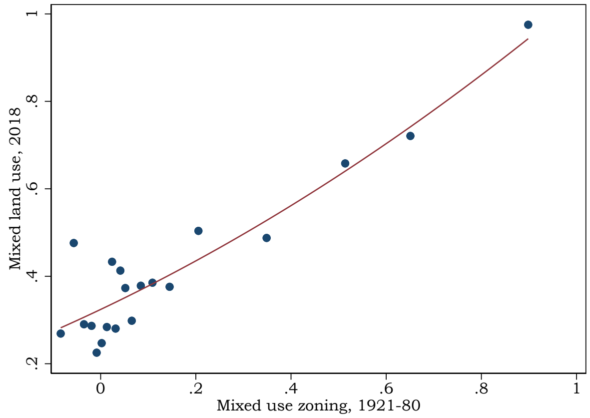
where y_i is our outcome of interest. Choices for y_i include: the diversity (entropy) index of land uses in 2018 (residential, townhouse, apartment, commercial, industrial) for quarter section i; the share of quarter section i occupied by apartments in 2018; the share of quarter section i occupied by commercial uses in 2018; and the average Walkscore of the quarter section in 2023. z_i is the diversity (entropy) index of zoning for the quarter section i for the earliest year for which we observe zoning covering at least half of the quarter section. Both present-day and historic entropy indices are standardized for comparability. The vector X includes controls for the quarter section's distance to the central business district of Chicago, distance to the boundary of Chicago, distance to the nearest of each zoning use type in Chicago in 1942 (industrial, commercial, apartment, townhome, and single family residential), distance to Lake Michigan, distance to the nearest major river, 1943 land value, and indicators for the median decade of development for parcels inside the square.

We restrict the sample to quarter sections that were at most 10% subdivided at the time they were first zoned so that we can assess the impact of regulation on the built environment using essentially undeveloped land, as depicted in Appendix Figure A2. We match the map of the zoning ordinance in place at the time of development. We begin with simple kernel regression estimations of the above equation in Figure 9. Figure 9a shows a the relationship between zoning diversity at the time of development and an index of mixed land uses today. We see a substantial upward trend, consistent with zoning having a substantial impact on the built environment over the long run. The rest of the figure shows sizable positive relationships between zoning diversity and the share of land set aside for commercial uses (Figure 9c), apartments (Figure 9b), and walkability (Figure 9d).

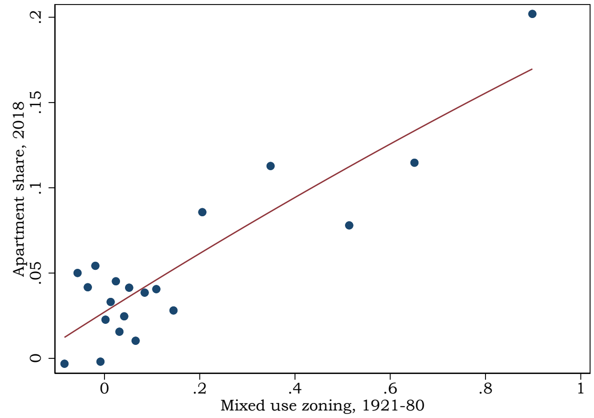
We also estimate the above equation and present the results in Table 4. A standard deviation increase in the diversity of zoning on land developed from 1921 to 1980 translates into a 0.7 standard deviation increase in real land use diversity today. We also see that more diverse initial zoning is associated with an increased share of land zoned for apartments and commercial uses, as well a reduction in the share of land zoned for single-family homes. A standard deviation reduction in zoning diversity at the time of development reduces the share of land set aside for commercial uses by 16% and the share of land set aside for apartments by 14%. These shifts come at the expense of single-family homes, whose land share drops by 45%.

Do these estimates reflect the zoning impact of zoning? In Table 4 we present the regression with each outcome with and without controls. The estimates are nearly invariant to the inclusion of controls for other factors that could drive these differences, including the location and value of the land. These results together suggest that zoning regulations themselves shifted the allocation

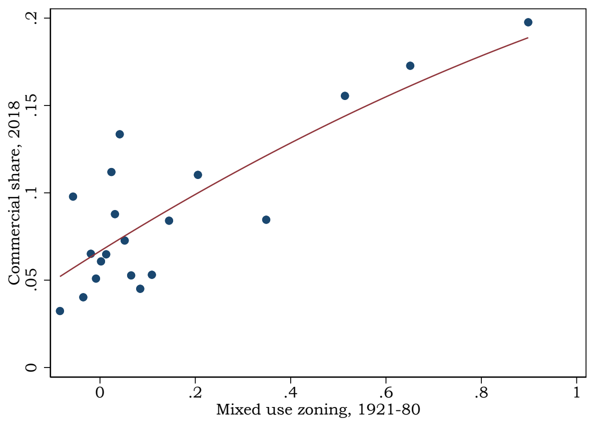
Figure 9: Impact of Diverse Zoning on Future Development



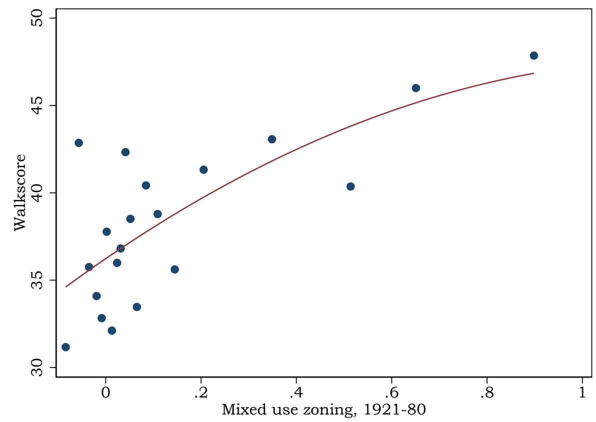
(a) Mixed land use index



(b) Share of land zoned for apartments



(c) Share of land zoned commercial



(d) Median Walkscore

Note: Binscatter plots with a quadratic fit. Outcome and predictor variables are orthogonalized with respect to controls for the quarter section's distance to the central business district of Chicago, distance to the boundary of Chicago, distance to the nearest of each zoning use type in Chicago in 1942 (industrial, commercial, apartment, townhome, and single family residential), distance to Lake Michigan, distance to the nearest major river, 1943 land value, and indicators for the median decade of development for parcels inside the square. The sample restricted to quarter sections that were at least 90% subdivided after zoning.

of land in the suburbs.

6 The Built Environment Today and Past Regulations

The data work in this paper required a years-long archival effort and the existence of an unusually well-equipped county GIS office. It would thus be extraordinarily difficult to replicate this approach for a broader sample of metro areas. Yet zoning has become central to many housing policy debates,

Table 4: Impact of Zoning on Land Use

	Land use diversity, 2018		Single family, 2018		Apartment, 2018		Commercial, 2018	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mixed use zoning, 1921-80	.71***	.71***	-.45***	-.47***	.14***	.14***	.16***	.15***
	(.05)	(.048)	(.048)	(.047)	(.028)	(.028)	(.031)	(.031)
Observations	334	334	334	334	334	334	334	334
R^2	0.39	0.32	0.42	0.21	0.22	0.12	0.20	0.12
Controls	Yes	No	Yes	No	Yes	No	Yes	No

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Note: Unit of observation is a quarter section and only the portion which is zoned. All regressions include covariates for the quarter section's distance to the CBD of Chicago, the boundary of Chicago, the nearest of each zoning use type in Chicago in 1942, Lake Michigan, and the nearest river, as well as the total area of the quarter section that is zoned and indicators for the median decade of development for parcels inside the quarter section. The sample is restricted to quarter sections where 90% of parcels were subdivided under a zoning law. HC3 standard errors are reported.

and researchers will continue to demand better data on these regulations. We close the paper by considering what we can learn from our setting, in which nearly everything was observed, to inform research on broader geographic contexts.

First, recall that the date that land was subdivided is rarely observed and often decades before houses were built on the resulting parcels. To illustrate why this is a challenge for identifying the impact of rezoning, we demonstrate how easy it is to "estimate" an economically meaningful effect of zoning when these regulations had little scope for impact. Consider our baseline estimate of the impact of mixed-use zoning on the diversity of the built environment today from the first column Table 4. The effect of .71 is identified from greenfield development, that is, quarter sections that were not subdivided before zoning was adopted. We repeat this estimate in the first column of Table 5. In column (2), we run the same regression on quarter sections that were subdivided when zoning was adopted by not yet built up. In these areas, zoning could still have had some scope for impact. The coefficient is .66 and significant at the one percent level.

We then restrict the sample to quarter sections where zoning could *not* have had much of an impact, namely because land was already developed. In column (3) we restrict attention to quarter sections that were subdivided before zoning was adopted in the municipality. Zoning still appears to have an economically important albeit attenuated impact on land use diversity today. In column (4) we consider only quarter sections that were entirely developed, that is, subdivided and built up, prior to the adoption of zoning. Even in these areas we "estimate" a weakly significant effect of zoning. But zoning had virtually no scope to impact the diversity of land use in these areas. These results underscore how difficult it is to credibly estimate effects of zoning when these regulations are almost always designed around existing development. In other contexts, researchers simply cannot observe whether regulation or land subdivision came first. We emphasize that "year built" information from the census or transaction records does not solve this problem because this date may be decades after the lots were drawn.

Table 5: Causal and "Causal" Effects of Zoning

	Land use diversity, 2018			
	(1)	(2)	(3)	(4)
Mixed use zoning, 1921-80	.71***	.66***	.45***	.32*
	(.05)	(.037)	(.051)	(.17)
	334	671	325	52
Controls	Yes	Yes	Yes	Yes
Subdivided after zoning	Yes	No	No	No
Built after zoning	No	Yes	No	No
Subdivided before zoning	No	No	Yes	No
Built before zoning	No	No	No	Yes

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Note: Unit of observation is a quarter section and only the portion which is zoned. All regressions include covariates for the quarter section's distance to the CBD of Chicago, the boundary of Chicago, the nearest of each zoning use type in Chicago in 1942, Lake Michigan, and the nearest river, as well as the total area of the quarter section that is zoned and indicators for the median decade of development for parcels inside the quarter section. HC3 standard errors are reported.

Because zoning is increasingly of interest to economists and policymakers, and many of the natural questions of interest involve regulations in place in the past, the literature has been substantially limited by the lack of information on historic zoning laws.

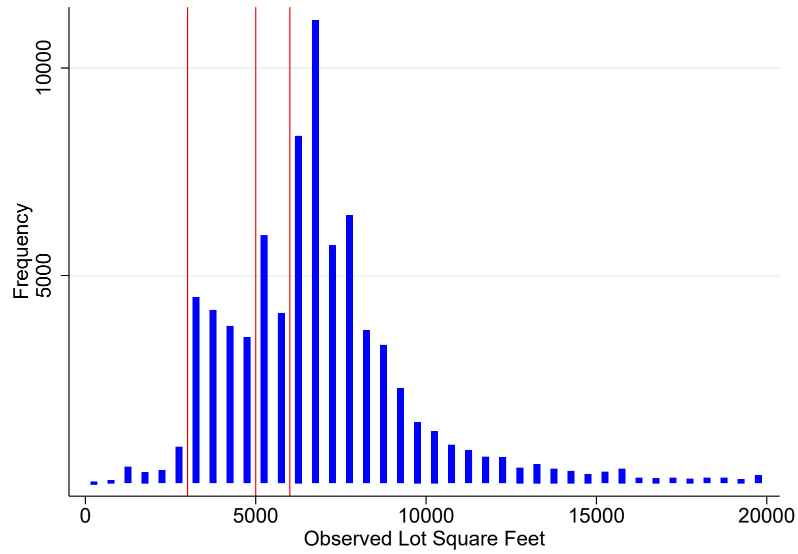
The difficulties of measuring timing manifest across approaches to estimate the impact of zoning. One increasingly common approach to measuring past lot size regulations without an archival data effort is to impute these zoning parameters from contemporary lot sizes and the date of construction of houses. Several recent papers use such approaches to consider the impact of minimum lot sizes on various outcomes such as home prices and racial segregation (Cui, 2024; Macek, 2024; Song, 2025; Zabel & Dalton, 2011). These papers estimate minimum lot size regulations using discontinuities in the distribution of lots sizes across some area observed in property tax records or other national datasets. The intuition is that lot sizes will "bunch" to the right of binding minimum lot sizes, making it possible to detect such regulations using econometric methods. What we can learn from our setting to guide interpretation of these more indirect approaches?

The primary difficulty with these approaches that we have very little understanding of what unregulated development would have looked like at the same time and place. Put differently, how lumpy would the lot size distribution be if private developers had subdivided the land before zoning ordinances were in place? An advantage of the dataset we have assembled for Cook county is that we can gain intuition on precisely this question as well as the distribution of lot sizes for neighborhoods platted before zoning. Because it is infeasible to reconstruct the parcel subdivision timeline and the complete history of maps and bylaws at a national level as we have done for Cook County, imputation methods will remain important for future work. Our goal here is to better understand what these approaches are measuring.

We begin with top panel of Figure 10, which shows the distribution of lot sizes for all homes built in the 1950s across incorporated suburbs of Cook County. We see a substantial degree of "bunching" to the right of common MLSs such as 3,000, 5,000, and 6,000 square feet. Do these jumps in the distribution reflect regulation? Since we can observe the date of subdivision and match it to the regulations in place at the time, we can repeat this exercise restricting to homes built on parcels that were subdivided *before* zoning was adopted in the municipality. Perhaps surprisingly, we see even more pronounced bunching around the same three minimum lot sizes. Because these lots were all subdivided before zoning was present, this bunching around round numbers is solely the result of developer choices.

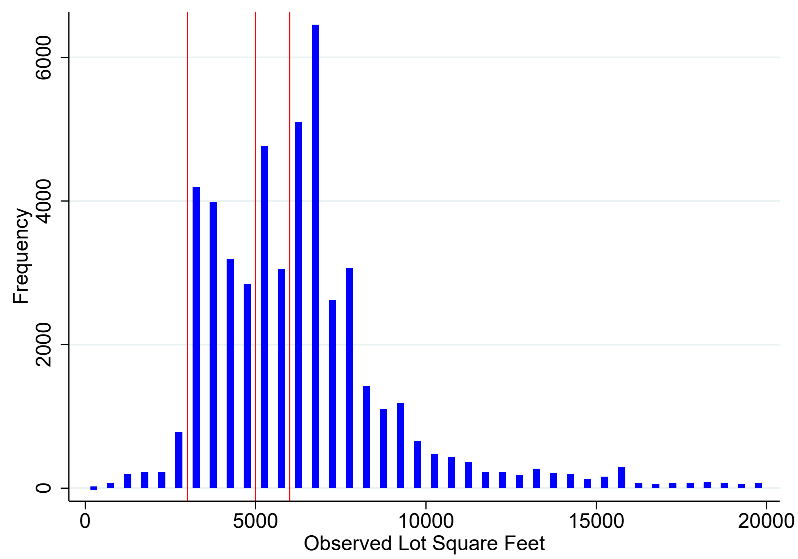
Why do lot sizes chosen by private developers exhibit this same bunching around round numbers? Advertisements in newspapers at the time show that developers often stated sizes of lots they were selling to aspiring homeowners. It was likely more appealing to subdivide land evenly and advertise "new homes on 5000 square foot lots!" rather than advertise lots of slightly smaller dimensions (appealing to the converse of the marketing logic that guides price setting). There is

Figure 10: The Distribution of Lot Sizes for Suburban Homes Built in 1950s



(a) All houses built 1950s

Note: N = 77,996 parcels built in the 1950s. Red lines represent 3 common MLS values = 3,000, 5,000, and 6,000.



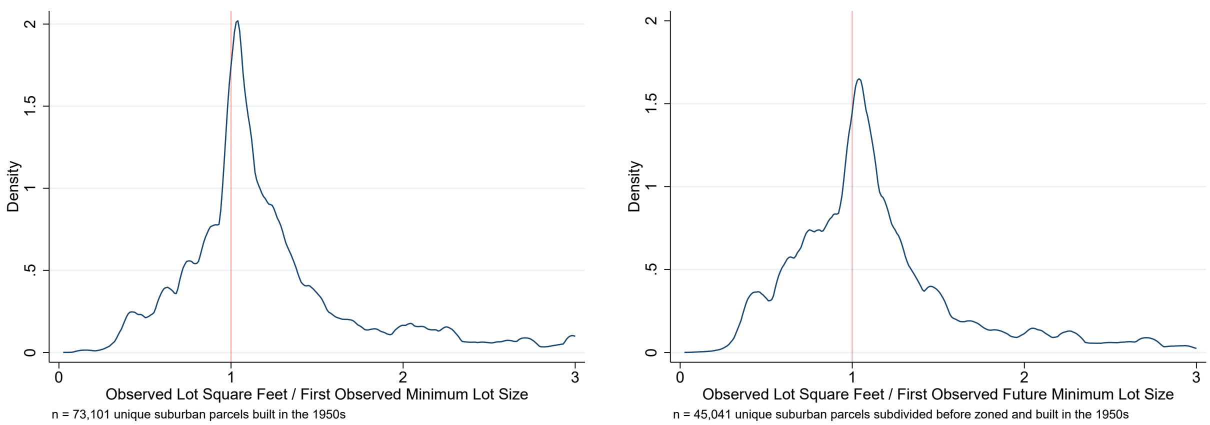
(b) Houses on lots subdivided before zoning

Note: N = 48,764 parcels subdivided before they were zoned and built in the 1950s. Red lines represent 3 common future MLS values = 3,000, 5,000, and 6,000.

also ample historical evidence to suggest that developers intended to market their developments to buyers of a certain class. Uniformly large lots were one of many tools used to exclude lower-income households, and particularly African-American households, from these communities.

If developer lot size choices bunch around common round numbers to a similar extent whether regulations were in place or not, what can economists learn about regulation from the current built environment? In the top panel of Figure 11, we plot the ratio of observed lot sizes to either the minimum lot size in place at the time, or the first future minimum lot size to be adopted for that parcel if it was subdivided before zoning, for all homes built in the 1950s. As expected, we see the most mass at 1, consistent with lot size regulations being binding. However, if we reproduce the same plot for only parcels that were subdivided before zoning was adopted and compute the ratio of lot sizes to the *future* MLS, we see almost the same degree of mass around 1 (right panel of Figure 11). This figure is consistent with what we know about historical land use regulation from the city of Chicago (Shertzer et al., 2018), which is that zoning often followed existing land use patterns.

Figure 11: Minimum Lot Sizes Followed Earlier Subdivision



(a) Built in 1950s, first observed MLS

(b) Subdivided before zoning, future MLS

Put differently, minimum lot sizes can appear binding either because they were, or because regulations were designed to match existing subdivision, which frequently featured lot sizes of particular round numbers. Although it is difficult to detect whether the lot size distribution is driven by regulation or private markets, they are an important part of the built environment and worthy of further study, particularly because they are very costly to change (Brooks & Lutz, 2016).

7 Conclusion

In this paper we construct the first panel dataset of zoning regulations and introduce the resulting Cook County Longitudinal Zoning Database. The dataset we have assembled for this paper represents the a new opportunity to investigate the impact of land use regulation on the development of suburbs. In particular, we demonstrate the importance of observing the timing of the various forces that operated on undeveloped land. Areas that were developed before zoning was adopted have smaller residential lots and more mixed uses relative to those developed under zoning. We argue that zoning drove much of this shift in the built environment and changed suburban form relative to what the market would have provided, even in the presence of strong demand for car-centric, low-density residential neighborhoods.

Zoning remains a critically important area of future research, although substantial obstacles have held back our understanding of the impact of these policies over the long run. Challenges for future work include developing methods to recover historical regulations related to allowable land uses, which to our knowledge has never been attempted. The results of this paper suggest that zoning was used to purge commercial uses and apartments from the suburbs, but this idea should be explored in other contexts beyond Chicago. A final challenge in this area is that wide gaps between the date a parcel was platted and the date a home was actually constructed may exist in many historical settings. The difficulty of observing subdivision dates on a large scale is an important consideration for research on lot sizes.

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A Appendix

A.1 Cook County Longitudinal Database Construction

The Cook County Longitudinal Database (CCLD) was construed from numerous public and private sources. Official records sourced directly from municipalities or academic and public libraries were used as our main source of data when available. Archives of local newspapers, which often printed zoning ordinances (including amendments) and zoning maps, were also an important source of official information, particularly for the earlier years of our study period. Another private source of official data was a three-volume (nearly comprehensive) collection of official suburban Cook County zoning ordinances as of May 1991/92 published by the Index Publishing Corporation. Information from these volumes was used in instances when a municipality's zoning ordinance in place as of 1991/92 reflected an older (unchanged) ordinance relevant to our study period. Alternative sources of detailed, yet unofficial, zoning data were used to populate the remainder of the database only in instances when official records were unavailable. Our primary sources of private zoning data were the various historic compendiums distributed by Geo. C. Olcott & Co. publishers, most notably editions of its *Olcott's Land Values Book of Chicago*, *Olcott's Land Values Blue Book of Cook County*, and *Chicago Zoning Ordinance and Zoning Maps* publications, each of which provides detailed information on select Cook County suburbs' zoning ordinances.

A suburb's zoning ordinance at any given period in time consists of two separate and equally-important components: (1) its *bylaws*, and (2) its *map*. For our purposes, an ordinance's bylaws make up the explicit language outlining use parameters for any given parcel of land withn the municipality's corporate boundaries. The aspects of an ordinance's bylaws most important to our study include:

- The establishment of individual and mutually exclusive zoning districts with the municipality
- The numerous land-use restrictions unique to each zoning district (e.g., allowed uses, minimum lot size requirements, height restrictions, etc.)
- The range of dates within which a particular set of bylaws' regulations were in effect

We worked to digitize key elements of each municipality's bylaws in a way that was both unified and temporally continuous over our study period. That is, we set out to construct a database that allowed us to observe the same set of particular zoning parameters on any given date (month/day/year) within our study period. This involved obtaining a copy of each municipality's first-ever zoning ordinance, digitizing its relevant bylaws (e.g., minimum lot requirement, etc.) and repeating the exercise for any subsequent known amendments (including but not limited to comprehensive amendments) that occurred up through or beyond our project's study period. The dates (and/or ordinance #) of particular amendments are (almost) always documented within the language of the bylaws themselves, thus allowing us to make specific public records requests to a municipality for copies of the official records when needed.

A zoning ordinance's zoning map allows us to observe how the numerous restrictions outlined within the bylaws vary across space at any particular point in time. However, maps can change shape even when the underlying bylaws do not; this is most often due to municipal annexation of new territory or individual parcels or tracts of land being rezoned upon request of individual land owners. The CCLD therefore relies on observing a series "snapshots" of a municipality's zoning map as it evolves over time in order to gauge the spatial evolution of its zoning bylaws. For municipalities first zoned in the 1920s, we attempt to digitize the zoning map as it would have looked at the time that the original (i.e., first) zoning ordinance passed. For all subsequent periods, we digitize the maps shape as it would have been in the years centered around 1940, 1950, 1960, and 1970.[12] We take further steps (discussed below) to better account for nuanced changes in the map's shape over continuous time, particularly as it relates to municipal annexation of new territory.

As with an ordinance's bylaws, we rely first on official zoning maps when available. Official maps were sourced from municipal records, academic and public libraries, and a handful of private sources such as newspaper archives and the Index Publishing Corporation's volume of scanned maps as of the 1991/92. The bylaws themselves also often provided official language describing a map's unique boundaries, particularly in earlier years when maps were less complex and therefore easy enough to describe using legal descriptions of land. Copies of unofficial boundaries were taken primarily from the Olcott's sources identified above.

Individual maps were digitized by overlaying the map (either digitally via georeferencing techniques or visually) onto GIS shapefiles outlining the contemporary locations of individual land parcels and streets/roads. This added shapefiles helped us identify nuanced spatial variation in zoning district boundaries that often followed individual parcel boundaries.

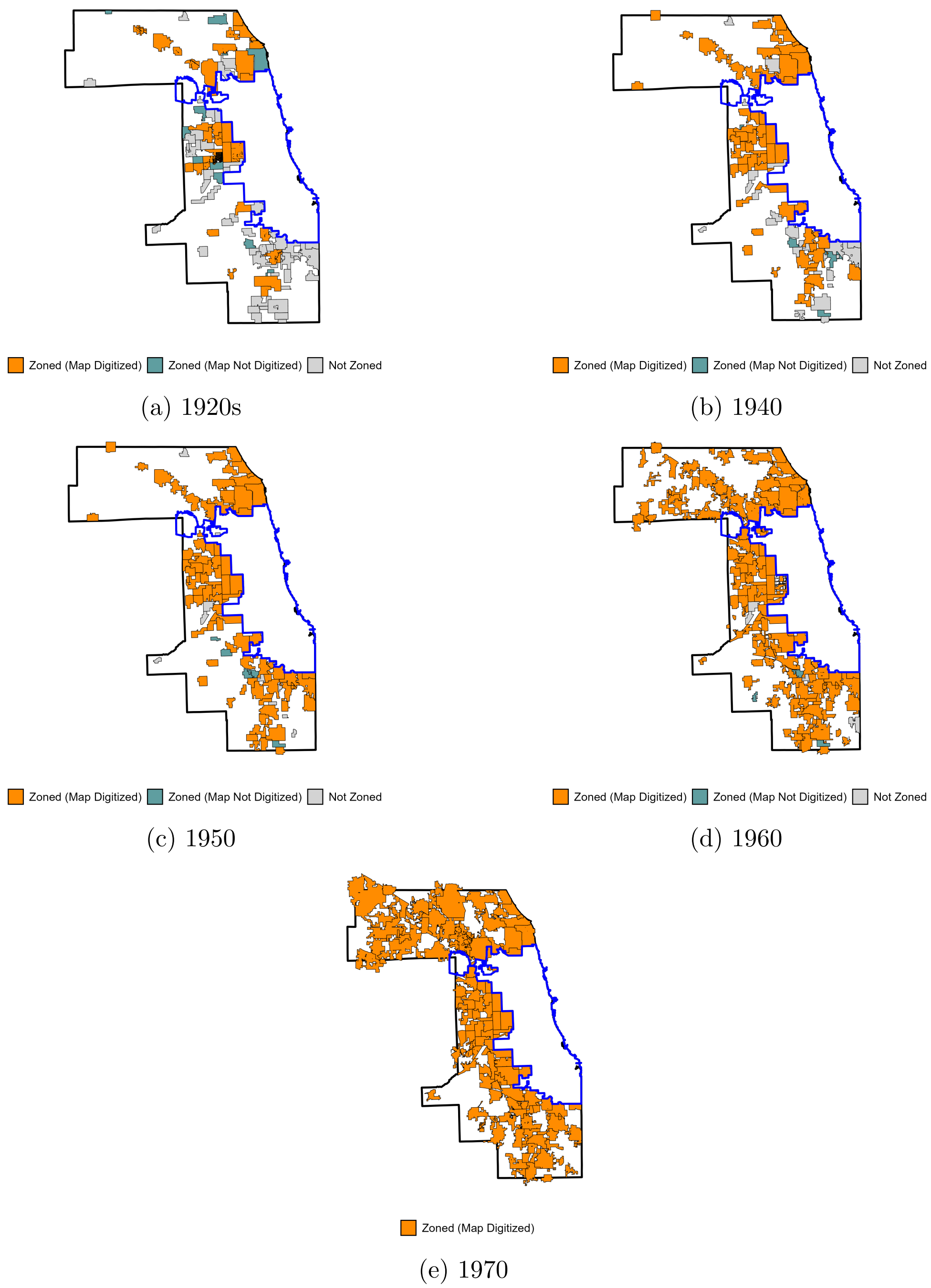
Special care was taken to assure that the maps we digitize for a specific time period (i.e., 1950, 1960, etc.) were relevant to the set of bylaws digitized for that same time period. For example, if a comprehensive amendment to a municipality's bylaws was passed in 1961, thus making those the bylaws we would have assigned to the 1960 bylaws file (because it falls within +/- 4 of 1960) we take steps to make sure that our 1960-era map to which these new bylaws are to be assigned reflects a period on/after the new bylaw's 1961 date of passage and not, for example, a 1959 map. Bylaws for an individual municipality's unique zoning districts are then assigned to their corresponding maps' unique zoning districts based on a common zoning district, municipality, and year identifiers.

Figure A1 depicts the municipal landscape for each of the five "snapshot" periods included in the CCLD. Suburban municipalities incorporated during each map's respective period are identified as being either "zoned" or "unzoned". Zoned municipalities are further stratified based on whether or not we were able to successfully digitize their zoning map for that particular period. The city of Chicago is outlined in blue within each map. The outer boundary (in black) outlines Cook County.

[12]We rely on source maps dated as close as possible to the year ending in '0' when digitizing maps for 1940/50/60/70. We typically allow for a window of time of no more than +/- 4 years (centered around 1940/50/60/70) when choosing source maps to digitize. For example, we made every attempt to use maps dated no earlier than 1964 and no later as 1956 to construct the municipality's "1960 map".

Altogether, these maps depict an expanding suburban landscape that was evenly split between "zoned" and "unzoned" in the 1920s but within which zoning became nearly universally adopted by 1950. The county's suburban land was almost fully covered by a zoned municipality' unique zoning regulations by the early 1970s.

Figure A1: Digitized Zoning Maps



Note: See text for details regarding how these were constructed.

A.2 Building a Parcel-Level Dataset

Data contained in the digitized zoning maps are assigned to individual suburban land parcels using a multi-step process involving numerous data files. We begin with a GIS shapefile of all Cook County land parcels as of 2017. This file is distributed by Cook County's GIS department and provides the exact location and shape for each parcel. We merge this GIS file with another file provided by the Cook County Assessor's office that provides the classification code and age for each parcel as of 2016.[13] This allows us to distinguish residential and nonresidential parcels as well as identify parcels used for single-family residential purposes.

The parcel file is also spatially merged with two additional publicly-provided shapefiles in order to assign each parcel its: (1) most recent date of subdivision, and (2) date of incorporation/annexation into whichever municipality it was located in (if any) at the time of subdivision.

The subdivision shapefile (along with its accompanying attribute file) was obtained through a public records request to Cook County's GIS office. This file identifies the shape and location of individual subdivision plats as they were in 2017 as well as the exact date the plat was officially recorded within the county clerk's office. We use this shapefile to assign the exact date of (the most recent) subdivision to each individual parcel.[14] A parcel's date of subdivision is of particular importance to our study because a zoning ordinance's language regulating lot size almost always references the date of subdivision as the key cutoff date for determining whether or not the associated regulations stipulated within the ordinance are binding or not. Parcels subdivided prior to the passage of a particular zoning ordinance are almost always "grandfathered in" within the ordinance's nonconforming use clause, thus allowing their size and shape to remain in place. However, parcels subdivided after the zoning ordinance's passage must comply with the new ordinance's lot size restrictions, among other regulations outlined within the ordinance's bylaws.

The second shapefile, Cook County's Municipal Incorporation Inventory (MII), is a unique resource that documents the (nearly-complete) history of municipal incorporation/annexation activity within the county. Just like private real estate subdividers, individual municipalities must file a plat of incorporation/annexation with the county clerk's office for official record-keeping purposes. The MII is the product of an ambitious undertaking whereby Cook County set out to digitize the complete history of these municipal plats. The MII, like the subdivision shapefile, identifies the shape and location of nearly all historical municipal incorporation/annexation plats as well as their official date of record. We spatially merge our parcel file to the MII in order to assign each parcel the date at which it was annexed to whichever municipality would have had authority to restrict its development at the time that it was subdivided (if it was under the authority of a particular municipality at the time that it was subdivided). [15]

[13]Detailed characteristics beyond assessor classification codes and age are available for residential properties but are not used in our analysis. A property's age is set to "NA" for open undeveloped land and set to "NA". Tax-exempt properties, which are often publicly owned, are dropped.

[14]We assign parcels their earliest-observed date of subdivision in the rare case that a single parcel overlaps multiple subdivision plat features.

[15]Unlike the subdivision shapefile, it is not uncommon for a single single parcel to overlap multiple municipal incorporation/annexation features contained within the MII shapefile. This is because the MII was built with an eye

Once each individual parcel is assigned its appropriate subdivision and incorporation/annexation dates, the full parcel file is merged with the full collection detailed zoning maps. This is done separately for each time frame for which we have a collection of digitized zoning maps (see Figure A1). We then append these files. In order to avoid any problems associated with small boundary errors in our digitized zoning maps, we keep only those parcels that, within a time frame, overlap zero or one zoning districts. We then determine the date that a parcel was zoned by interacting the date that it was incorporated/annexed into its municipality with the date the municipality itself was first zoned. The algorithm is straightforward:

- If date(muni zoned) $<$ date(parcel inc/anx into muni) $\Rightarrow$ date(parcel zoned) $=$ date(parcel inc/anx into muni)
- If date(muni zoned) $>$ date(parcel inc/anx into muni) $\Rightarrow$ date(parcel zoned) $=$ date(muni zoned)
- Parcel observations overlapping zero zoning districts for a particular time frame receive "NA" for date(parcel zoned)

We then separate the parcel file into three mutually exclusive sub-files based on whether or not they were zoned at the time of subdivision and whether or not we observe the detailed map/bylaws if zoned at the time of subdivision.

A.3 Table of Maps and Access Locations

towards documenting the complete history of municipal boundary changes over time which often leads to overlapping plats across different periods of time. For example, it was not uncommon for a piece of land to be annexed to one municipality, disconnected from that municipality at a later date, and then re-annexed into another municipality at an even later date. Timelines such as these introduce considerable complexity to the structure of the data once the MII and parcel file are spatially merged. The steps outlined below were taken in order to assign to each parcel the inc/anx date that would be most appropriate for determining which municipality's zoning ordinance (if any) would have been binding at the time that the parcel was subdivided. (Assigning the appropriate zoning ordinance occurs in a later step.)

For parcels overlapping only a single MII feature:

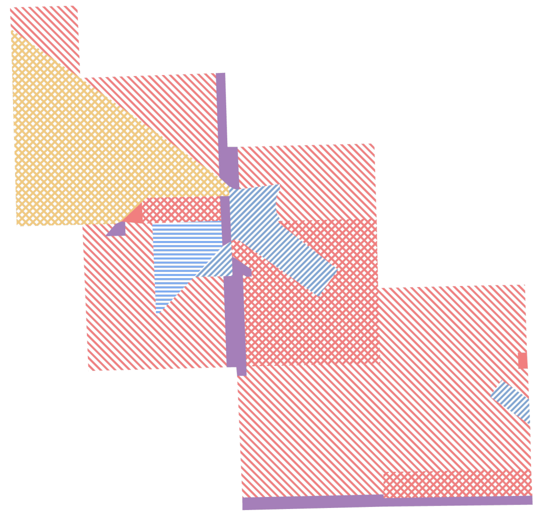
1. Assign the inc/anx date associated with that particular MII feature.

For parcels overlapping multiple MII features:

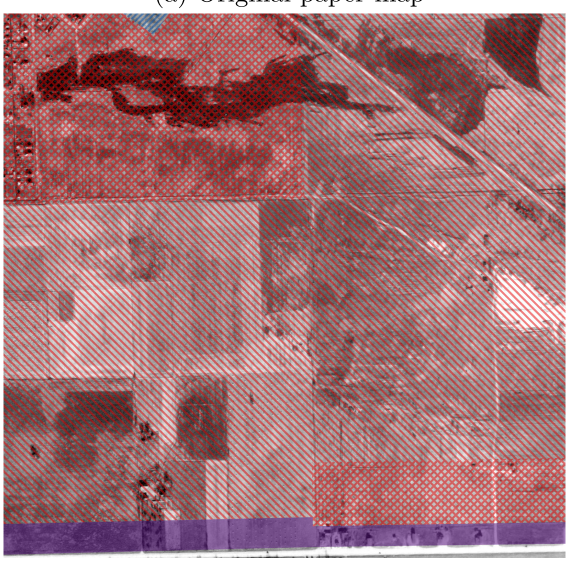
1. If the subdivision date precedes *all* overlapping inc/anx dates, then assign the latest inc/anx date *out of convenience*. This parcel was not subject to any municipal zoning ordinance at the time of subdivision so the date that it was incorporated/annexed is irrelevant to assigning any zoning ordinance that would have been in place at the time of subdivision.
2. If the subdivision date precedes all but one inc/anx date, then assign the latter as the inc/anx date. This particular inc/anx date (and the associated municipality) is the only one that would have been associated with a potentially-binding municipal zoning ordinance at the time of subdivision.
3. If the subdivision date comes after multiple inc/anx dates, assign the latest of these inc/anx dates, as this inc/anx date is most likely aligned with whichever municipality the parcel belonged to at the time that it was subdivided. These parcels reflect cases where they initially belonged to one municipality, were then disconnected from that municipality, and then re-annexed into another municipality (after which they were eventually subdivided, perhaps under the latter municipality's binding zoning ordinance if one was in place at the time).

Figure A2: Zoning Maps for Chicago Ridge, IL in 1945

(a) Original paper map



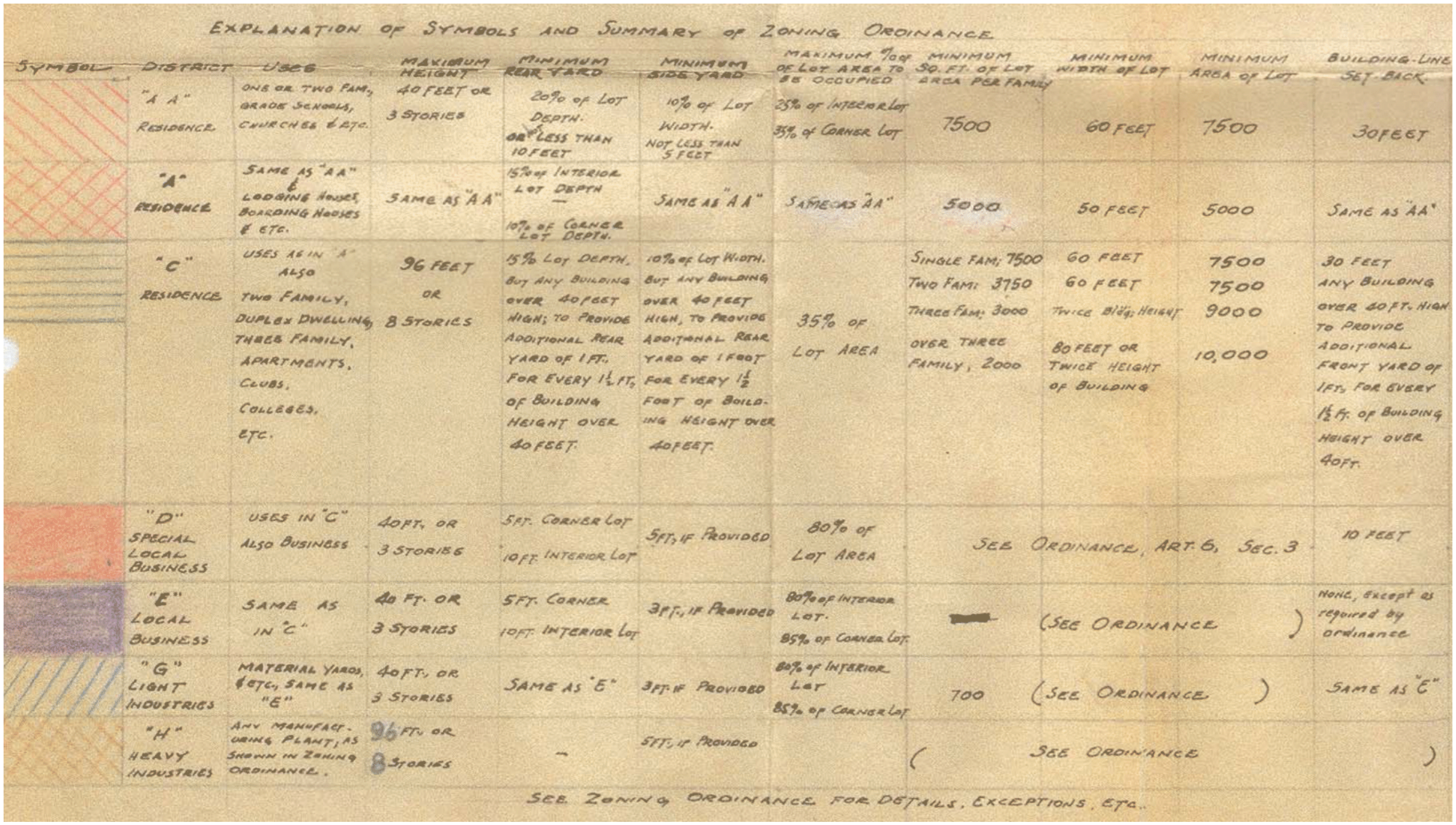
(b) Digitized zoning map



(c) Zoning variation in 1945

(d) Aerial survey photo, 1938-9

Figure A3: Chicago Ridge 1945 Ordinance Legend



EXPLANATION OF SYMBOLS AND SUMMARY OF ZONING ORDINANCE

SYMBOL	DISTRICT	USES	MAXIMUM HEIGHT	MINIMUM REAR YARD	MINIMUM SIDE YARD	MAXIMUM % OF LOT AREA TO BE OCCUPIED	MINIMUM SQ. FT. OF LOT AREA PER FAMILY	MINIMUM WIDTH OF LOT	MINIMUM AREA OF LOT	BUILDING-LINE SET-BACK
	"A A" RESIDENCE	ONE OR TWO FAM., GRADE SCHOOLS, CHURCHES & ETC.	40 FEET OR 3 STORIES	20% OF LOT DEPTH. OR LESS THAN 10 FEET	10% OF LOT WIDTH. NOT LESS THAN 5 FEET	25% OF INTERIOR LOT 35% OF CORNER LOT	7500	60 FEET	7500	30 FEET
	"A" RESIDENCE	SAME AS "AA" & LODGING HOUSES, BOARDING HOUSES & ETC.	SAME AS "A A"	15% OF INTERIOR LOT DEPTH — 10% OF CORNER LOT DEPTH.	SAME AS "A A"	SAME AS "AA"	5000	50 FEET	5000	SAME AS "AA"
	"C" RESIDENCE	USES AS IN "A" ALSO TWO FAMILY, DUPLEX DWELLING, THREE FAMILY, APARTMENTS, CLUBS, COLLEGES, ETC.	96 FEET OR 8 STORIES	15% LOT DEPTH, BUT ANY BUILDING OVER 40 FEET HIGH; TO PROVIDE ADDITIONAL REAR YARD OF 1 FT., FOR EVERY 1½ FT, OF BUILDING HEIGHT OVER 40 FEET.	10% OF LOT WIDTH. BUT ANY BUILDING OVER 40 FEET HIGH, TO PROVIDE ADDITIONAL REAR YARD OF 1 FOOT FOR EVERY 1½ FOOT OF BUILDING HEIGHT OVER 40 FEET.	35% OF LOT AREA	SINGLE FAM; 7500 TWO FAM: 3750 THREE FAM: 3000 OVER THREE FAMILY, 2000	60 FEET 60 FEET TWICE BLDG. HEIGHT 80 FEET OR TWICE HEIGHT OF BUILDING	7500 7500 9000 10,000	30 FEET ANY BUILDING OVER 40 FT. HIGH TO PROVIDE ADDITIONAL FRONT YARD OF 1 FT., FOR EVERY 1½ FT. OF BUILDING HEIGHT OVER 40 FT.
	"D" SPECIAL LOCAL BUSINESS	USES IN "C" ALSO BUSINESS	40 FT. OR 3 STORIES	5 FT. CORNER LOT 10 FT. INTERIOR LOT	5 FT., IF PROVIDED	80% OF LOT AREA	SEE ORDINANCE, ART. 6, SEC. 3			10 FEET
	"E" LOCAL BUSINESS	SAME AS IN "C"	40 FT. OR 3 STORIES	5 FT. CORNER 10 FT. INTERIOR LOT	3 FT., IF PROVIDED	80% OF INTERIOR LOT. 85% OF CORNER LOT.	—	(SEE ORDINANCE	)	NONE, except as required by ordinance
	"G" LIGHT INDUSTRIES	MATERIAL YARDS, & ETC., SAME AS "E"	40 FT. OR 3 STORIES	SAME AS "E"	3 FT. IF PROVIDED	80% OF INTERIOR LOT 85% OF CORNER LOT	700	(SEE ORDINANCE	)	SAME AS "E"
	"H" HEAVY INDUSTRIES	ANY MANUFACTURING PLANT, AS SHOWN IN ZONING ORDINANCE.	96 FT. OR 8 STORIES	—	5 FT., IF PROVIDED		(	SEE ORDINANCE		)

SEE ZONING ORDINANCE FOR DETAILS, EXCEPTIONS, ETC.

Contains bylaws for each zoning district in 1945.